Sarvajith Guddety

ai24btech11008@iith.ac.in | sarvajithguddety@gmail.com | +91 7331159987 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.38
Narayana Junior College	Class XII (Board of Intermediate Education Andhra Pradesh)	2024	89.20%
Montessori EM Primary School	Class X (Central Board of Secondary Education)	2022	95.20%

SKILLS

- **Programming**: Python, C++, SQL
- **Machine Learning**: Scikit-learn, XGBoost, LightGBM, CatBoost, Model Development, Feature Engineering, Model Evaluation
- **Deep Learning**: PyTorch, TensorFlow, Keras, CNNs, RNNs, LSTMs, Transfer Learning, Hugging Face Transformers, RAG
- **Data Science**: NumPy, Pandas, SciPy, Matplotlib, Seaborn, Statistical Analysis, Data Visualization
- **Databases & Vector Stores**: SQL, FAISS
- **Tools**: Git, GitHub, Jupyter Notebook, Linux

RELEVANT COURSES

Mathematics: Calculus I, Calculus II, Vector Calculus, Differential Equations, Series of Functions, Matrix Theory, Optimization-I, Optimization-II, Introduction to Metric Spaces, Numerical Analysis

Machine Learning & AI: Foundations of Machine Learning, Deep Learning, Programming for AI, AI for Humanity

Computer Science: Discrete Maths for Computer Science, Data Structures & Applications, Algorithms, Database Management Systems 1

Statistics & Data Science: Probability and Random Variables, Concentration Inequalities, Introduction to Statistics

PROJECTS

Fair Bayesian Engine -- Fairness-Aware Active Learning for Credit Risk [GitHub]

Tech Stack: Python, NumPy, SciPy, scikit-learn, Matplotlib

- Developed a Fair Active Learning system combining Bayesian logistic regression (Laplace approximation) with fairness-aware sampling for credit-risk modelling on the Give Me Some Credit dataset.
- Designed and evaluated 4 label-selection strategies, increasing minority-group (age) representation from ~10% to ~50% over 150 active-learning steps without sacrificing accuracy, AUC-ROC, or F1.
- Implemented Balance Gradient Descent (5x minority sample weighting) and Monte Carlo-estimated fairness-gain scoring, substantially shrinking demographic parity and TPR/FPR gaps between age groups.

Industrial Surface Defect Detection with Explainable AI [GitHub]

Tech Stack: PyTorch, torchvision, Grad-CAM

- Built and benchmarked two PyTorch CNN architectures (a custom 4-block CNN vs. a fine-tuned EfficientNet-B0) for steel surface defect classification, reaching 98.15% and 100% test accuracy respectively across 6 defect classes.
- Implemented Grad-CAM from scratch using forward/backward hooks to visualize model attention, making predictions auditable for factory-floor deployment.
- Engineered a reproducible stratified 70/15/15 data pipeline with class-balanced augmentation and a full evaluation suite (precision, recall, F1, confusion matrix).

Patent Semantic Search Engine [GitHub]

Tech Stack: Python, Sentence-Transformers, FAISS, Hugging Face

- Built a semantic search engine for patents using Sentence-Transformers bi-encoder embeddings (all-MiniLM-L6-v2) indexed with FAISS for fast approximate nearest-neighbor retrieval.
- Implemented a Cross-Encoder reranking stage (ms-marco-MiniLM-L-6-v2) on top of retrieval to improve ranking precision on a BigPatent dataset subset.
- Achieved Recall@10 of 0.75, MRR of 0.59, and nDCG@10 of 0.98 on a held-out patent query evaluation set.
- Designed an end-to-end indexing, retrieval, reranking, and evaluation pipeline using only free, open-source models with no paid APIs.

SPSCQueue -- Ultra-Low-Latency Lock-Free Ring Buffer [GitHub]

Tech Stack: C++20, CMake, Google Benchmark, ThreadSanitizer

- Built a production-grade, wait-free single-producer/single-consumer lock-free ring buffer for ultra-low-latency inter-thread communication in financial systems.
- Implemented cache-line-aware layout (read/write cursors padded to separate 64-byte cache lines), a fixed-capacity raw aligned byte store with zero heap allocation on the hot path, C++20 concept-enforced power-of-two capacity, and perfect-forwarding in-place construction supporting move-only types.
- Achieved 262.4 million messages/sec throughput at 3.81 ns average latency across 10 million messages with zero data corruption; verified correct acquire-release semantics under ThreadSanitizer and zero false sharing via perf c2c.

EXTRACURRICULAR

Member, National Sports Organisation (NSO) - Badminton